IOT(Internet of Things)PROJECT DOCUMENTATION

Academic Year: 2024-2028
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Group Number: Group 12

1. Project Title

IoT Based Smart Refrigerator Theft Alert System

2. Group Details

Sr No	Student Name	Roll No	Contribution
1	Sameer	150096724117	Hardware Connections, Coding, Blynk App & Project Infra
2	Ragini	150096724023	Ideation
3	Saumya	150096724108	Documentation & Testing

4	Aditya	15009672 4062	Testing

3. Problem Statement

In hostels and shared offices, refrigerator doors are sometimes opened by unauthorized persons which leads to food theft. It becomes difficult to monitor who opened the refrigerator. Therefore, there is a need for a smart monitoring system that can detect when someone tries to open the refrigerator and send a notification to the user through a mobile application.

4. Objective

- To detect refrigerator door opening using IR sensor
- To send notification to the user using Blynk mobile app
- To activate buzzer alert during restricted hours
- To implement an IoT based monitoring system using ESP8266

5. Components Required

Component	Quantity	Purpose
ESP8266 NodeMCU	1	Main microcontroller
IR Sensor	1	Detects door

		movements
Buzzer	1	Provides alert sound
Jumper Wires	Few	Wiring connections
Blynk Mobile App	1	Send notifications to user
WiFi Network	1	Internet connection

6. Working Principle

The ESP8266 connects to Wi-Fi and communicates with the Blynk IoT platform. An IR sensor is placed near the refrigerator handle. The detection distance is adjusted using the potentiometer so that the handle alone is not detected.

When a person tries to open the refrigerator, their finger comes close to the IR sensor. The sensor detects the object and sends a signal to the ESP8266. The microcontroller then checks whether the current time is within the restricted hours.

If the access happens during restricted hours, the system activates a buzzer and sends a notification to the user through the Blynk mobile application.

7. Circuit Connections

ESP8266 Pin	Connected To
D2	IR Sensor OUT
3V	IR Sensor VCC
GND	IR Sensor GND
D5	Buzzer Positive
GND	Buzzer Negative

8. Program Code

```
#define BLYNK_PRINT Serial

#define BLYNK_TEMPLATE_ID ""
#define BLYNK_TEMPLATE_NAME ""
#define BLYNK_AUTH_TOKEN ""

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <NTPClient.h>
#include <WiFiUdp.h>

char ssid[] = "YOUR_WIFI_NAME";
char pass[] = "YOUR_WIFI_PASSWORD";

int sensor = D2;
int buzzer = D5;
```

```
WiFiUDP udp;
NTPClient timeClient(udp, "pool.ntp.org", 19800);

int startHour = 13;
int endHour   = 14;

void setup()
{
  Serial.begin(9600);

  pinMode(sensor, INPUT);
  pinMode(buzzer, OUTPUT);

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timeClient.begin();
}

void loop()
{
  Blynk.run();
  timeClient.update();

  int detect = digitalRead(sensor);
  int hour = timeClient.getHours();

  if(detect == LOW && hour >= startHour && hour <= endHour)
  {
     digitalWrite(buzzer, HIGH);
```

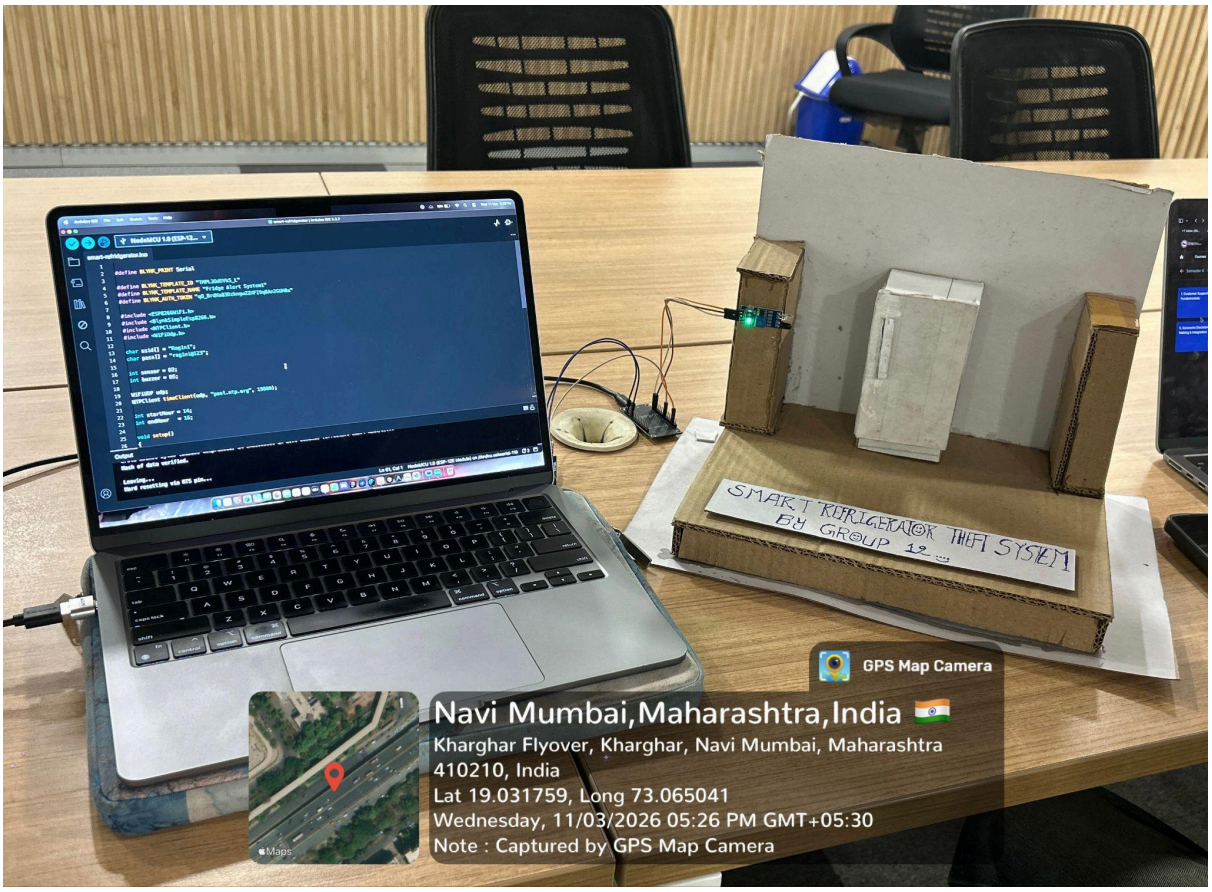
```
    Blynk.logEvent("fridge_open","Someone is opening the
fridge!");

    Blynk.virtualWrite(V0,"Door Opened");

    delay(3000);

    digitalWrite(buzzer, LOW);
  }
  else
  {
    Blynk.virtualWrite(V0,"Door Closed");
  }
}
```

9. Mobile Application (Blynk Setup)

- Install Blynk IoT mobile application
- Create new template for ESP8266 device
- Add device to the template
- Create event named "fridge_open"
- Enable notification for this event
- Add Label widget in dashboard
- Set Datastream to Virtual Pin V0

- Set Label name as "Fridge Status"
- Copy Auth Token and paste it in Arduino code
- When sensor detects object → ESP8266 sends data → Blynk shows status "Door Opened" and sends notification

10. Geo-Tagged Image of Project



11. Project Demonstration Video Link

Upload project demo video on Google Drive or YouTube (Unlisted)

Paste link here:

group 12.mp4

12. Output

Action	Result
Finger near sensor	IR sensor detects object
During restricted hours	Buzzer alert activates
System connected to Blynk	Mobile notification received

13. Applications

- Security monitoring in hostels
- Office refrigerator monitoring
- Shared kitchen access control
- Food theft prevention system

14. Advantages

- Low cost IoT solution
- Easy to build and implement
- Real-time mobile notification
- Helps monitor unauthorized access

15. Limitations

- Requires Wi-Fi connection

- Depends on power supply
- Sensor detection range must be adjusted properly

16. Future Scope

- Add camera monitoring system
- Add fingerprint or RFID authentication
- Add door open logging system
- Add temperature monitoring inside refrigerator

17. Conclusion

The Smart Refrigerator Theft Alert System successfully detects when someone tries to open the refrigerator using an IR sensor. The ESP8266 processes the signal and sends an alert through the Blynk IoT platform during restricted hours. This project demonstrates a simple and effective implementation of IoT technology for security monitoring in shared environments.